

Static assessments leave critical strategic value unoptimized

The Rationale: Why PULSE is needed now

TODAY

✗ Point estimates

Single scores (1-5) mask uncertainty — no confidence intervals

✗ Static snapshot

Excel V12 is updated annually — market moves weekly

✗ Forces treated independently

No causal propagation — regulatory shock $\neq$ tech shock

✗ No competitive response

Pool shifts modeled in isolation from competitor actions

✗ Subjective scoring

Unstructured scoring process introduces systematic bias



PULSE

✓ Bayesian distributions

Full probability distributions with 80% confidence intervals

✓ Continuous path modeling

Annual paths 2026-2030 with velocity tracking & triggers

✓ Causal DAG propagation

Shocks propagate through directed causal network with lags

✓ Game-theoretic responses

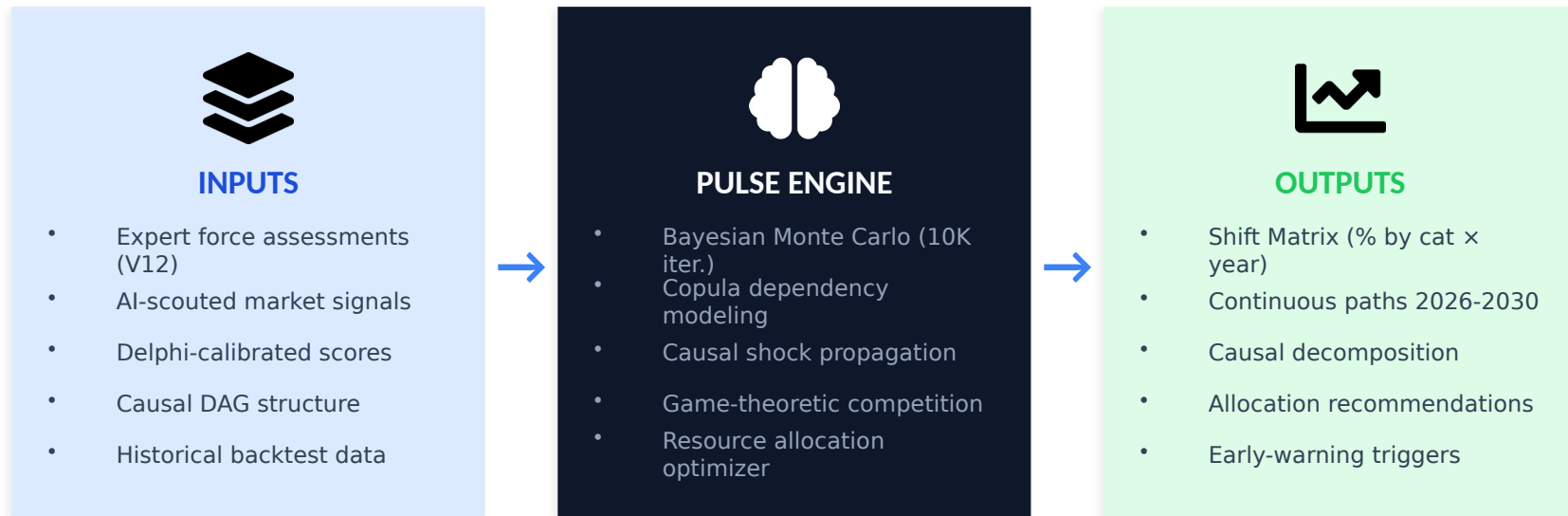
Competitor archetypes model second-order pool effects

✓ Delphi-calibrated scoring

Structured multi-round protocol with debiasing

PULSE transforms expert judgment into probabilistic profit pool intelligence

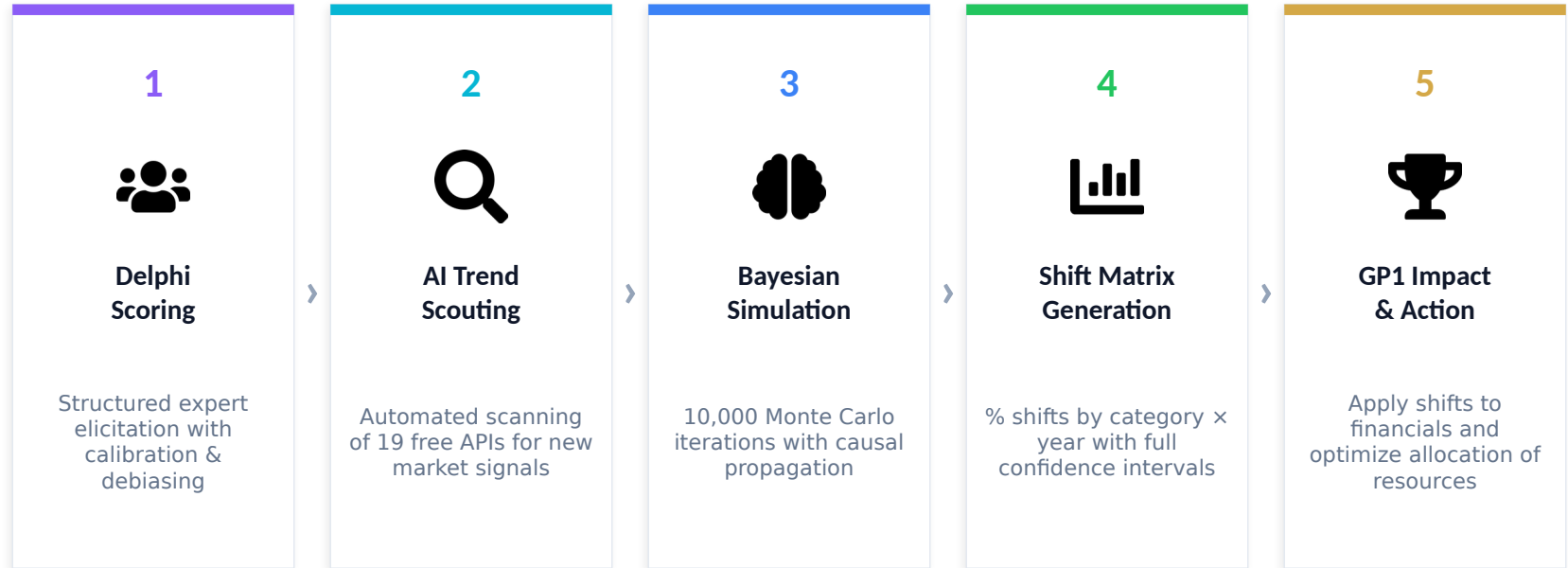
The Concept: A simulation engine, not a dashboard



Core Principle: PULSE outputs relative shifts only (%). Users apply shifts to their own financials. PULSE never sees €M figures.

Five-step workflow from expert scoring to strategic action

The Process: How teams interact with PULSE



Quarterly Rhythm: Q1 Calibration → Q2 Full Simulation → Q3 Mid-Year Refresh → Q4 Budget Integration

Delphi protocol replaces unstructured scoring with calibrated consensus

Deep Dive: Expert Elicitation with Debiasing



Quality Gates

Inter-Rater Reliability

$\alpha > 0.67$ required to proceed

Calibration Accuracy

Score vs. known historical outcomes

Bias Detection

Anchoring, optimism, recency flags

Convergence

Scores must stabilize within 3 rounds

Example: 'Natural / Clean Beauty' trend drives -3.2% Color pool shift by 2030

In Action: From trend scoring to category impact

CONSUMER FORCE › Natural / Clean Beauty Movement

Impact



4/5

Probability



4/5

Contraction

Bayesian Posterior

In pact ~ Beta(6,3) | Prob ~ Beta(6,3) | Delphia = 0.74

Category Exposure (0-5)

Color



Care



Styling



Body



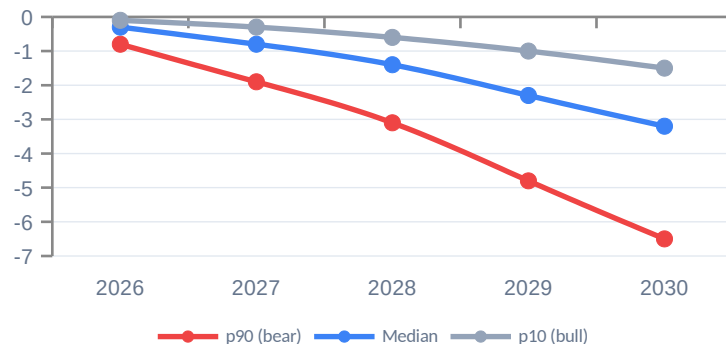
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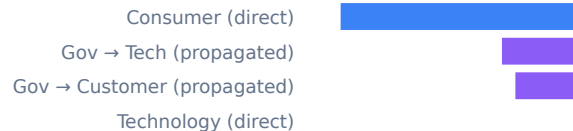
FCA



Hair: Color — Shift Path 2026-2030



Causal Decomposition



Modular Python + React architecture with provider-agnostic AI layer

Technology: Built for enterprise deployment within Henkel's cloud

INTERFACE LAYER

Export

React War Room (localhost:3000) | Excel Bridge (Import/Export) | REST API (FastAPI) | Power BI



AI LAYER (Provider-Agnostic)

Azure OpenAI

Trend Scanner | Score Calibrator | Scenario Narrator | NL Chat | MVP: Claude → Prod:



SIMULATION ENGINE

Bayesian MC
+ Copulas

Causal DAG
Propagation

Game Theory
Competitive

Resource
Optimizer

Continuous
Paths

Delphi
Protocol

Python 3.11+ | Num Py | SciPy | pandas | NetworkX | FastAPI | SQLite

DATA: SQLite | Excel V12 | Audit Log

EXTERNAL APIs (19, all free): GDELT GNews ECHA Google Trends ...

Three-phase rollout from CLI MVP to full enterprise deployment

Timeline: 12-month roadmap with clear team responsibilities

Phase 1

Bayesian MC + Causal Engine • Months 1-4

- CLI reads V12 → outputs Shift Matrix
- Bayesian MC with copula dependencies
- Causal DAG with shock propagation
- Deterministic V12 parity validation
- Backtesting engine (if history avail.)

Phase 2

War Room Dashboard • Months 5-8

- React War Room (single-view)
- Interactive heatmap + path timeline
- Causal flow animation
- Real-time re-simulation on changes
- Export center (Excel, PPTX, PDF)

Phase 3

AI Intelligence + Power BI • Months 9-12

- AI trend scanner (19 free APIs)
- Score calibration + narratives
- NL chat interface
- Delphi workflow automation
- Power BI semantic model export

REQUIREMENTS FROM TEAMS

Internal Tech / IT Team

- ☐ Python 3.11+ environment provisioning
- ☐ Azure OpenAI deployment (Henkel tenant)
- ☐ Power BI workspace + semantic model setup
- ☐ InfoSec review for API integrations
- ☐ SSO / identity integration

Category / Strategy Teams

- ☐ Participate in Delphi scoring rounds (3x/yr)
- ☐ Provide V12 Excel + historical versions
- ☐ Review & validate AI-suggested trends
- ☐ Define early-warning trigger thresholds
- ☐ Champion adoption within category teams

Executive Summary

PULSE transforms Henkel's static force assessment into a living, probabilistic strategic decision platform — delivering Bain-grade profit pool intelligence at zero external cost.



Bayesian Rigor

Full probability distributions with copula dependencies and backtested credibility scores



Causal Intelligence

Directed causal network propagates shocks — each force type has a unique signature



Competitive Awareness

Game-theoretic layer models competitor responses and second-order pool effects



Decision-Ready Outputs

Allocation optimizer, early-warning triggers, and continuous paths drive strategic action



Zero Running Cost

19 free APIs, local Python engine, provider-agnostic AI — no recurring license fees



Enterprise-Grade

Azure OpenAI in Henkel tenant, Power BI integration, audit trail — built for ExCo trust

Recommendation: Approve Phase 1 build (4 months) to deliver backtested CLI with Shift Matrix output